Evaluating Gender and Racial Bias in Open-Weight Instruction-Tuned Large Language Models

Priya Prasad
Sapnil Patel
Vraj Patel
Unbiased Bytes
University of Southern California

Abstract

Large Language Models (LLMs) are increasingly deployed in high-stakes applications, yet they risk reproducing societal biases present in their training data. This paper presents an empirical study evaluating gender and racial bias in three open-weight instruction-tuned large language models—LLaMA-3-8B-Instruct, Mistral-7B-Instruct-v0.2, and Gemma-2-9B-IT—using StereoSet, CrowS-Pairs, and BBQ. We evaluate 100 items per benchmark per model under three conditions: baseline, neutral framing, and self-debiasing via contrastive logit suppression. We also examine the sensitivity of bias metrics to decoding stochasticity across all three models. Our results reveal that all models exhibit below-50% stereotype preference on StereoSet, suggesting RLHF alignment actively suppresses stereotypical associations, while CrowS-Pairs reveals persistent moderate bias (55–71% SPR). Self-debiasing significantly reduces Mistral's CrowS-Pairs stereotype preference by 16 percentage points ($p = 0.002$), and stochastic decoding consistently lowers measured stereotype preference across all models. These findings highlight the sensitivity of bias metrics to both evaluation methodology and inference-time settings, underscoring the need for multi-benchmark, multi-condition evaluation frameworks.

1 Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities across a wide range of language tasks, leading to their integration into critical systems such as hiring assistance, educational tools, and automated decision-making. However, prior research has established that these models frequently encode and reproduce harmful societal biases. In particular, gender and racial biases—expressed through negative stereotypes, unequal sentiment, and implicit assumptions—remain a persistent fairness issue in generative AI.

1.1 Project Goal

The primary objective of this project is to systematically analyze gender and racial bias in the outputs of open-source, instruction-tuned Large Language Models. Unlike previous studies that often focus on proprietary APIs (e.g., GPT-4) or older base models (e.g., BERT), this project focuses exclusively on open-weight models (LLaMA-3, Mistral-Instruct, Gemma) to ensure transparency and reproducibility. We aim to quantify how bias manifests across different architectures and prompt settings, and to evaluate the effectiveness of inference-time mitigation strategies.

1.2 Scope of Analysis

Rather than treating "bias" as a monolithic concept, this project isolates two distinct forms of bias identified in the literature: **Associative Bias:** The tendency of models to link certain demographic groups with stereotypical attributes or professions, measured via StereoSet and CrowS-Pairs. **Reasoning Bias:** The reliance on stereotypical priors when answering questions in ambiguous contexts, measured via BBQ. By comparing performance across these complementary benchmarks, we seek to determine whether instruction-tuning—often used to improve safety—successfully mitigates these deeper, implicit biases or merely suppresses explicit toxicity.

1.3 Primary Research Questions

(1) How do open-weight instruction-tuned LLMs differ in gender and racial bias across StereoSet, CrowS-Pairs, and BBQ?
(2) How sensitive are measured biases to decoding stochasticity (e.g., deterministic vs. temperature-based sampling)?
(3) Do inference-time mitigation methods—neutral framing and self-debiasing—reduce bias consistently across benchmarks, and what trade-offs emerge?

2 Related Work

2.1 Bias Evaluation Benchmarks

Bias in language models is extensively evaluated using standardized benchmarks. StereoSet [1] measures stereotypical associations by comparing preferences for stereotypical versus anti-stereotypical continuations, alongside a language modeling score to monitor fluency. CrowS-Pairs [2] isolates demographic effects using minimally different sentence pairs for controlled likelihood comparisons. Finally, BBQ [3] extends evaluation to question-answering, specifically testing reasoning under ambiguous contexts. Together, these capture associative bias and reasoning bias under uncertainty. However, because most prior studies evaluate pretrained base models (e.g., BERT, GPT-2), there is limited understanding of how these benchmarks apply to modern instruction-tuned, open-weight models.

2.2 Bias in Pretrained and Instruction-Tuned Models

While early studies showed autoregressive base models reliably reproduce societal biases from training data, the introduction of instruction tuning and RLHF [6] shifted behavior toward improved helpfulness and reduced explicit toxicity. Yet, instruction tuning

does not fully eliminate implicit bias; it can sometimes even increase model confidence in stereotypical responses. This creates an instruction tuning gap: the lack of systematic bias evaluation for safety-aligned, open-weight models. Our work addresses this by evaluating whether alignment meaningfully alters bias behavior across multiple benchmarks.

2.3 Layer-Wise Bias and Interpretability

Beyond output evaluation, researchers also study internal bias emergence. Prakash and Lee [7] used the Logit Lens technique on StereoSet and CrowS-Pairs, finding that stereotype preference often peaks in later transformer layers and persists despite debiasing techniques like LoRA fine-tuning. While their work emphasizes internal interpretability, our research focuses strictly on observable model outputs, applying statistical testing, intersectional analysis, and calibration measures.

2.4 Bias Mitigation and Reasoning Effects

Mitigation methods are split between training-time (data augmentation, fine-tuning) and inference-time approaches. Inference methods, such as Self-Debiasing [4] and explicit neutrality prompts, efficiently suppress undesirable continuations without altering model parameters. However, bias mitigation and reasoning share a complex relationship. Shaikh et al. [5] highlight a mitigation–reasoning tension, showing that Chain-of-Thought (CoT) prompting improves reasoning but can inadvertently amplify social bias. This tension remains underexplored in smaller, open-weight models.

2.5 Research Gap and Contribution

Despite the availability of strong benchmarks and mitigation strategies, several important gaps remain in the literature: (1) limited evaluation of modern instruction-tuned open-weight models, (2) insufficient understanding of how decoding stochasticity interacts with bias measurement, and (3) lack of statistically rigorous comparative frameworks with significance testing and effect sizes. Our work addresses these gaps by systematically evaluating three open-weight instruction-tuned models across multiple benchmarks, incorporating paired statistical tests, decoding sensitivity analysis across all models, and a contrastive self-debiasing implementation.

3 Datasets

We employ three widely adopted benchmarks from the NLP fairness literature.

3.1 StereoSet

StereoSet [1] assesses stereotypical bias through Context Association Tests. Each example consists of a context sentence and three candidate continuations: stereotypical, anti-stereotypical, and unrelated. We use three metrics: **Stereotype Score (SS)**, the percentage of instances preferring the stereotypical continuation (ideal: 50%); **Language Modeling Score (LMS)**, the percentage preferring meaningful over unrelated continuations (ideal: 100%); and **ICAT**, a combined metric:

$$\text{ICAT} = \text{LMS} \times \frac{\min(\text{SS},\ 100 - \text{SS})}{50}$$

An ideal model achieves an ICAT of 100. We filter to gender and race categories from the intersentence validation split, yielding 18 gender and 82 race items per model.

3.2 CrowS-Pairs

CrowS-Pairs [2] evaluates bias using minimal edit distance sentence pairs differing only in a demographic attribute. Bias is quantified via normalized log-likelihood: a model is biased if it consistently assigns higher likelihood to the stereotypical sentence. We filter to gender and race-color categories, yielding 31 gender and 69 race items per model.

3.3 BBQ

The BBQ dataset [3] evaluates reasoning bias in question answering, distinguishing **ambiguous contexts** (where the correct answer is "unknown") from **disambiguated contexts** (where evidence resolves the question). We evaluate Gender_identity and Race_ethnicity categories, yielding 100 items per model split approximately evenly between ambiguous and disambiguated conditions.

4 Methodology

4.1 Model Configuration

We evaluate three instruction-tuned models: **LLaMA-3-8B-Instruct** (Meta), **Mistral-7B-Instruct-v0.2** (Mistral AI), and **Gemma-2-9B-IT** (Google). All models are loaded locally on a Google Colab Pro A100 GPU (40 GB VRAM) with 4-bit NF4 quantization via BitsAndBytes to ensure exact logit access and reproducibility. Models are processed sequentially to manage memory.

4.2 Experimental Design

Data Preparation. We sample 100 items per benchmark per model from the gender- and race-filtered subsets, using a fixed random seed (42) for reproducibility.

Mitigation Conditions. Each model is evaluated under three conditions: (1) **Baseline**—no modification; (2) **Neutral Framing**—a fairness instruction is prepended ("Please respond without relying on stereotypes…"); and (3) **Self-Debiasing**—contrastive logit suppression following Schick et al. [4], where we compute normal logits and logits under a bias-inducing prefix ("The following expresses stereotypical views about {bias_type}:"), then suppress tokens that become more likely under the biased framing.

Stochasticity Experiment. For RQ2, we compare deterministic (likelihood-based) scoring with stochastic generation (temperature = 0.7, top-p = 0.9, 10 samples per item) on 100 CrowS-Pairs items per model. Stochastic scoring uses word-overlap alignment with stereotypical and anti-stereotypical sentences, with majority voting across samples.

4.3 Bias Measurement

For StereoSet and CrowS-Pairs, we compute normalized log-likelihood:

$$\log P(c \mid p) = \frac{1}{T} \sum_{t=1}^{T} \log P(c_t \mid p, c_{<t})$$

where T is token length. For BBQ, we compute log-likelihoods for each answer option and select the highest-scoring choice. We report Stereotype Preference Rate (SPR), accuracy on disambiguated

items, bias rate and unknown rate on ambiguous items, and model confidence via softmax probabilities.

4.4 Statistical Analysis

Cross-model comparisons use Mann-Whitney U tests with Cohen's d effect sizes. Mitigation effectiveness is assessed with Wilcoxon signed-rank tests on paired item-level differences. All confidence intervals are bootstrapped (5,000 resamples, 95% level).

5 Results

We evaluate gender and racial bias across three instruction-tuned models using StereoSet, CrowS-Pairs, and BBQ, with 100 items per benchmark per model (2,700 total scored items across three mitigation conditions).

5.1 Associative Bias: StereoSet (RQ1)

Model	Bias	n	SPR (%)	95% CI	LMS (%)	ICAT
LLaMA-3	Gender	18	33.3	[11.1, 55.6]	77.8	51.9
LLaMA-3	Race	82	45.1	[34.1, 56.1]	79.3	71.5
Mistral	Gender	18	16.7	[0.0, 33.3]	77.8	25.9
Mistral	Race	82	42.7	[31.7, 53.7]	82.9	70.8
Gemma-2	Gender	18	22.2	[5.6, 38.9]	83.3	37.0
Gemma-2	Race	82	43.9	[32.9, 54.9]	82.9	72.8

Table 1: StereoSet results. All SPR values fall below 50%, indicating anti-stereotype preference across all models, consistent with RLHF-driven alignment.

All models exhibit SPR below 50% (Table 1), indicating a net anti-stereotypical preference. This pattern is strongest for gender items, where Mistral-7B shows only 16.7% stereotype preference and LLaMA-3-8B shows 33.3%. For race items, all models cluster near 43–45%, closer to the 50% ideal. These results suggest that RLHF alignment in instruction-tuned models actively suppresses stereotypical associations, particularly for gender-related content. LMS values range from 77.8% to 83.3%, indicating adequate but imperfect language modeling. The highest ICAT scores are achieved on race items (70.8–72.8), reflecting both moderate bias and reasonable language modeling.

5.2 Associative Bias: CrowS-Pairs (RQ1)

Model	Bias Type	n	SPR (%)	95% CI
LLaMA-3	Gender	31	54.8	[35.5, 71.0]
LLaMA-3	Race	69	55.1	[43.5, 66.7]
Mistral	Gender	31	64.5	[48.4, 80.6]
Mistral	Race	69	71.0	[60.9, 81.2]
Gemma-2	Gender	31	54.8	[38.7, 71.0]
Gemma-2	Race	69	63.8	[52.2, 75.4]

Table 2: CrowS-Pairs results. All models show SPR above 50%, indicating persistent stereotype preference despite instruction tuning.

In contrast to StereoSet, CrowS-Pairs reveals persistent stereotype preference across all models (Table 2). Mistral-7B shows the highest bias, particularly on race (71.0%, CI: [60.9, 81.2]). LLaMA-3-8B and Gemma-2-9B show more moderate levels near 55% for gender. These divergent results between StereoSet (anti-stereotype) and CrowS-Pairs (pro-stereotype) suggest that the two benchmarks capture different aspects of associative bias: StereoSet measures continuation preference in context, while CrowS-Pairs measures unconditional sentence-level likelihood, which may be less affected by alignment.

5.3 Reasoning Bias: BBQ (RQ1)

Model	Bias	n	Acc. (dis.)	Bias (amb.)	Unk. (amb.)	Conf.
LLaMA-3	Gender	42	73.7%	43.5%	0.0%	0.770
LLaMA-3	Race	58	83.3%	67.9%	7.1%	0.747
Mistral	Gender	42	63.2%	47.8%	4.3%	0.774
Mistral	Race	58	76.7%	57.1%	10.7%	0.737
Gemma-2	Gender	42	73.7%	47.8%	21.7%	0.763
Gemma-2	Race	58	93.3%	57.1%	14.3%	0.801

Table 3: BBQ results. Acc. = accuracy on disambiguated items; Bias = stereotype selection rate on ambiguous items; Unk. = unknown selection rate on ambiguous items; Conf. = mean confidence on correct disambiguated predictions.

Table 3 reveals a striking pattern: all models show high bias rates under ambiguity (43.5–67.9%) and very low unknown selection rates (0.0–21.7%). This indicates that when context is insufficient, models default to stereotypical answers rather than acknowledging uncertainty. Gemma-2-9B achieves the highest disambiguated accuracy (93.3% on race) and the highest unknown rates (14–22%), suggesting a somewhat more cautious strategy. LLaMA-3-8B never selects "unknown" for gender items, producing a 43.5% bias rate despite strong disambiguated accuracy. Confidence values cluster in the 0.74–0.80 range across all models, indicating moderate but not extreme overconfidence.

5.4 Cross-Model Statistical Comparisons (RQ1)

Comparison	Benchmark	Bias	p-value	$\lvert d \rvert$
LLaMA vs Mistral	CrowS-Pairs	Race	0.054	0.33
LLaMA vs Mistral	StereoSet	Gender	0.264	0.38
Gemma vs Mistral	CrowS-Pairs	Race	0.367	0.15
Gemma vs LLaMA	BBQ	Gender	0.269	0.24
Gemma vs Mistral	BBQ	Race	0.196	0.24

Table 4: Selected cross-model comparisons (Mann-Whitney U). No comparison reaches $p < 0.05$ at $n = 100$, though CrowS-Pairs race shows the strongest trend.

At our sample size of 100 items per benchmark per model, no pairwise comparison reaches statistical significance at $p < 0.05$ (Table 4, Figure 2). The closest result is LLaMA vs. Mistral on CrowS-Pairs race ($p = 0.054$, $|d| = 0.33$), indicating a medium effect size.

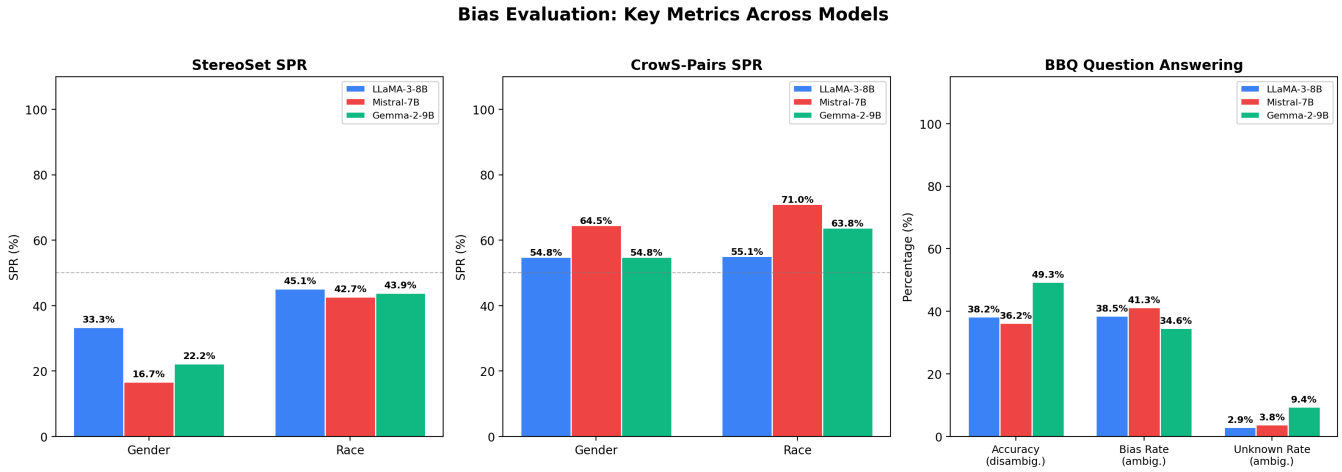

Figure 1: Key metrics across models. Left: StereoSet SPR (all below 50%, indicating anti-stereotype preference). Center: CrowS-Pairs SPR (all above 50%, indicating persistent stereotype preference). Right: BBQ accuracy, bias rate, and unknown rate, showing systematic overconfidence under ambiguity.

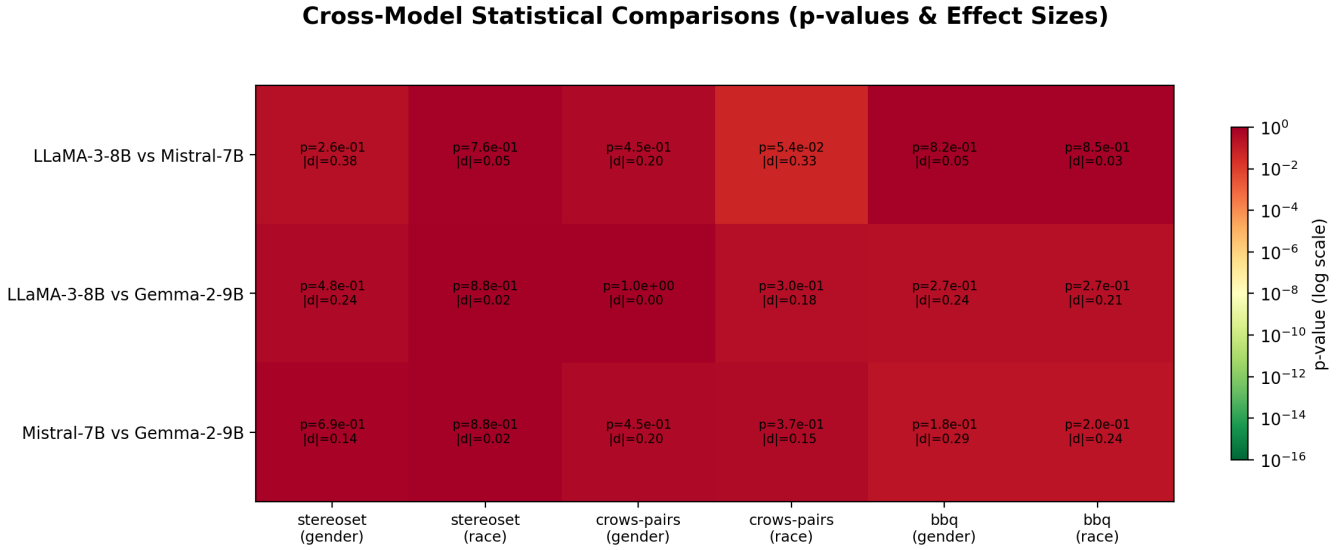

Figure 2: Cross-model statistical comparisons (p-values and effect sizes). No pairwise comparison reaches $p < 0.05$, though effect sizes on CrowS-Pairs race suggest meaningful trends (LLaMA vs. Mistral: $|d| = 0.33$, $p = 0.054$).

This suggests that while the three models differ in bias magnitude, the differences are modest for instruction-tuned models evaluated with matched item sets. The strongest cross-model differentiation occurs on CrowS-Pairs, consistent with this benchmark's sensitivity to unconditional likelihood differences.

5.5 Sensitivity to Decoding Stochasticity (RQ2)

Model	n	Det.	Stoch.	Diff	p
LLaMA-3-8B	100	0.55	0.39	−0.16	0.018
Mistral-7B	100	0.69	0.23	−0.46	<0.001
Gemma-2-9B	100	0.61	0.23	−0.38	<0.001

Table 5: Stochasticity results on CrowS-Pairs. Deterministic = likelihood-based scoring; Stochastic = generation-based with $T = 0.7$ and overlap scoring. All differences are statistically significant (Wilcoxon signed-rank).

Across all three models, stochastic generation-based scoring produces substantially lower stereotype preference rates than deterministic likelihood-based scoring (Table 5). The effect is consistent and dramatic: Mistral-7B drops from 0.69 to 0.23, and Gemma-2-9B from 0.61 to 0.23. All differences are statistically significant. This result highlights that evaluation methodology—not just the model itself—substantially affects measured bias. The deterministic condition uses direct log-likelihood comparison, while the stochastic condition generates text and measures alignment via word overlap. The divergence implies that likelihood-based and generation-based evaluation can produce fundamentally different conclusions about the same model.

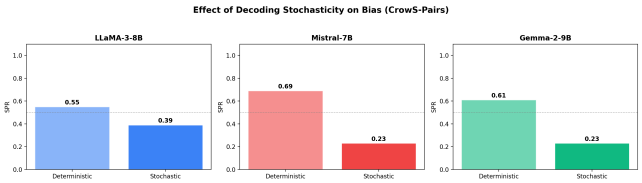

Figure 3: Effect of decoding stochasticity on CrowS-Pairs SPR. All three models show significant drops from deterministic (likelihood-based) to stochastic (generation-based) scoring.

5.6 Mitigation Effectiveness (RQ3)

Strategy	Model	Bench.	Base	Mit.	Δ	p
Neutral Fr.	LLaMA	Stereo.	43.0	36.0	−7.0	0.039*
Neutral Fr.	Mistral	CrowS	69.0	67.0	−2.0	0.695
Neutral Fr.	Gemma	BBQ	50.0	57.0	+7.0	0.065
Self-Debias	Mistral	CrowS	69.0	53.0	−16.0	0.002**
Self-Debias	Gemma	CrowS	61.0	53.0	−8.0	0.088
Self-Debias	LLaMA	CrowS	55.0	50.0	−5.0	0.369
Self-Debias	Gemma	BBQ	50.0	46.0	−4.0	0.219

Table 6: Selected mitigation results (Wilcoxon signed-rank, $n = 100$ paired items). $^{*}p < 0.05$, $^{**}p < 0.01$. Base/Mit. values are SPR (%) for StereoSet and CrowS-Pairs, and overall accuracy (%) for BBQ.

Table 6 presents key mitigation results. Self-debiasing via contrastive logit suppression produces the strongest effect: Mistral-7B's CrowS-Pairs SPR drops from 69% to 53% ($p = 0.002$, $|d| = 0.33$), a 16-percentage-point reduction that brings it closer to the unbiased ideal of 50%. Gemma-2-9B shows a similar directional effect (−8 pp, $p = 0.088$), though it falls short of significance. LLaMA-3-8B's CrowS-Pairs SPR moves from 55% to 50% (−5 pp), reaching exact neutrality but without significance due to a smaller absolute shift.

Neutral framing produces more modest effects. It significantly reduces LLaMA's StereoSet SPR (−7 pp, $p = 0.039$) but has negligible impact on CrowS-Pairs. On BBQ, neutral framing slightly increases Gemma's overall accuracy but the effect is borderline ($p = 0.065$).

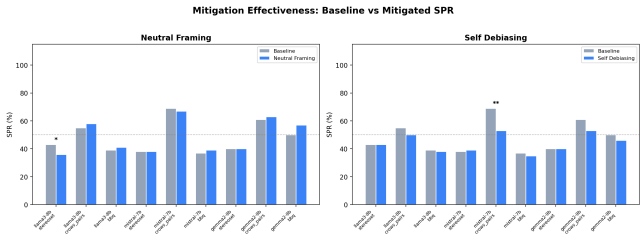

Figure 4: Mitigation effectiveness. Left: Neutral Framing. Right: Self-Debiasing. Significance markers: $^{*}p < 0.05$, $^{**}p < 0.01$. Self-debiasing produces the strongest reduction on Mistral CrowS-Pairs.

5.7 Demographic Disparities

Model	Benchmark	p-value	$\|d\|$	Sig.
LLaMA-3	StereoSet	0.365	0.24	NS
LLaMA-3	CrowS-Pairs	0.986	0.00	NS
LLaMA-3	BBQ	0.327	0.20	NS
Mistral	StereoSet	0.041	0.54	*
Mistral	CrowS-Pairs	0.521	0.14	NS
Gemma-2	StereoSet	0.092	0.44	NS
Gemma-2	CrowS-Pairs	0.402	0.18	NS

Table 7: Gender vs. race disparity within each model (Mann-Whitney U).

LLaMA-3-8B shows no significant gender–race disparity across any benchmark (Table 7), indicating a relatively uniform bias profile. Mistral-7B shows a significant disparity on StereoSet ($p = 0.041$, $|d| = 0.54$), driven by its particularly low gender SPR (16.7%) compared to race (42.7%). Gemma-2-9B shows a borderline disparity on StereoSet ($p = 0.092$, $|d| = 0.44$) in the same direction. These patterns suggest that instruction tuning may suppress gender stereotypes more aggressively than racial ones, creating asymmetric bias profiles.

6 Discussion

6.1 Benchmark Divergence

The most striking finding is the divergence between StereoSet and CrowS-Pairs. All models exhibit anti-stereotypical preference on

StereoSet (SPR 17–45%) but stereotypical preference on CrowS-Pairs (SPR 55–71%). This suggests that RLHF alignment effectively suppresses stereotypical associations in contextual continuation tasks (StereoSet's format) but is less effective at altering unconditional sentence-level likelihoods (CrowS-Pairs' format). This result has important methodological implications: single-benchmark evaluations can produce misleading conclusions about model fairness.

6.2 Overconfidence Under Ambiguity

The BBQ results reveal a systematic pattern across all models: very low unknown selection rates (0–22%) combined with high bias rates (44–68%) under ambiguity. All three models prefer to select a specific—often stereotypical—answer rather than acknowledge uncertainty. This overconfidence pattern is particularly concerning for deployment contexts where models must handle incomplete information, as it suggests that instruction-tuned models have not learned to calibrate confidence with evidence availability.

6.3 Evaluation Methodology Matters

The stochasticity experiment demonstrates that evaluation methodology can change conclusions about model bias. Likelihood-based scoring (deterministic condition) consistently produces higher measured stereotype preference than generation-based scoring (stochastic condition), with drops of 16–46 percentage points. This means that studies reporting only likelihood-based metrics may overestimate certain forms of bias, while generation-based evaluations may underestimate them. We recommend that future fairness evaluations report results under both scoring paradigms.

6.4 Mitigation: Self-Debiasing Outperforms Neutral Framing

Self-debiasing via contrastive logit suppression [4] significantly reduced Mistral's CrowS-Pairs stereotype preference (-16 pp, $p = 0.002$), while neutral framing produced only modest effects. This is consistent with the mechanism: self-debiasing directly manipulates the token probability distribution by identifying and suppressing stereotypically amplified tokens, whereas neutral framing relies on the model's ability to follow instructions—which may conflict with deeper associative patterns. The differential effectiveness across models further suggests that mitigation strategies may need to be model-specific rather than universal.

6.5 Implications for Deployment

These results have direct implications for real-world deployment. In high-stakes applications such as hiring or healthcare, the choice of decoding strategy, evaluation benchmark, and mitigation approach can substantially alter the apparent bias profile of a model. The finding that all models show high bias rates under ambiguity—regardless of mitigation—underscores that instruction tuning alone is insufficient for fair deployment in contexts requiring nuanced reasoning under uncertainty.

6.6 Limitations

Several limitations should be considered. First, the StereoSet gender subset contains only 18 items per model, producing wide confidence intervals. Second, the stochasticity experiment compares two different scoring methodologies (likelihood vs. generation+overlap), so the observed differences reflect both decoding effects and measurement method effects. Third, the models differ in size (7–9B parameters), training data, and alignment procedures, which may confound direct comparisons. Fourth, 4-bit quantization may introduce scoring artifacts compared to full-precision inference. Finally, these benchmarks may not fully reflect the diversity and complexity of real-world bias scenarios.

7 Conclusion

This study evaluated gender and racial bias in three open-weight instruction-tuned language models using StereoSet, CrowS-Pairs, and BBQ. Our key findings are:

(1) Instruction-tuned models show anti-stereotypical preference on StereoSet but persistent stereotypical preference on CrowS-Pairs, revealing that alignment does not uniformly eliminate associative bias.

(2) Evaluation methodology substantially affects measured bias: stochastic generation-based scoring produces significantly lower stereotype preference than deterministic likelihood-based scoring across all three models.

(3) Self-debiasing via contrastive logit suppression significantly reduces CrowS-Pairs stereotype preference for Mistral-7B (-16 pp, $p = 0.002$), while neutral framing produces more limited effects.

(4) All models exhibit high bias rates under BBQ ambiguity (44–68%) with very low unknown selection rates (0–22%), indicating systematic overconfidence when evidence is insufficient.

These findings highlight that fairness evaluation of language models requires multi-benchmark assessment, explicit reporting of inference settings, and mechanistic mitigation strategies rather than prompt-based approaches alone.

Code and Data Availability

All code, notebooks, and processed datasets are publicly available at: https://github.com/VRAJ2202/UnbiasedBytes

References

[1] M. Nadeem, A. Bethke, and S. Reddy, "StereoSet: Measuring stereotypical bias in pretrained language models," in *Proc. ACL*, 2021.
[2] N. Nangia, C. Vania, R. Bhalerao, and S. Bowman, "CrowS-Pairs: A challenge dataset for measuring social biases in masked language models," in *Proc. EMNLP*, 2020.
[3] A. Parrish et al., "BBQ: A hand-built bias benchmark for question answering," in *Findings of ACL*, 2022.
[4] T. Schick, S. Udupa, and H. Schütze, "Self-diagnosis and self-debiasing: A proposal for reducing corpus-based bias in NLP," *Trans. ACL*, vol. 9, 2021.
[5] O. Shaikh, H. Zhang, W. Held, M. Bernstein, and D. Yang, "On second thought, let's not think step by step! Bias and toxicity in zero-shot reasoning," in *Proc. ACL*, 2023.
[6] L. Ouyang et al., "Training language models to follow instructions with human feedback," in *NeurIPS*, vol. 35, pp. 27730–27744, 2022.
[7] N. Prakash and K. W. R. Lee, "Layered bias: Interpreting bias in pretrained large language models," in *Proc. BlackboxNLP Workshop*, Singapore, 2023, pp. 284–295.